Data Analysis Programming — Semester 2026-II
Universidad Distrital Francisco José de Caldas · School of Engineering

Extreme Weather Monitoring System — Texas

Workshop No. 1 — Hazard Definition, Source Validation, and Data Contract
Project Definition Report · Team 5

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November 2026

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1. Project Overview

Title. Extreme Weather Monitoring System — Texas.

Motivation. Texas concentrates almost every type of extreme weather recorded in the United States: heat waves, hard freezes (the February 2021 winter storm), flash floods, large hail, tornadoes, damaging thunderstorm winds, dust storms and winter storms. In 2025 alone the NOAA Storm Events Database lists 5,641 Texas records. The information needed to judge whether today's weather is unusual is public, but it is spread over services with different formats, time conventions, identifiers and update rhythms: bulk CSV archives that are recompiled months later, a daily climate archive with a three-day delay, and real-time APIs that only keep a few days of data.

Goal. From a *data-engineering* perspective: build a reproducible pipeline that loads 2010–present history from two bulk/historical sources, polls two operational NWS services incrementally with watermarks, and integrates the four into one canonical PostgreSQL/PostGIS model with validation, quarantine, deduplication and cross-source event grouping. From an *analytical* perspective: compare each current measurement with the historical distribution of the same station and season and classify it as **NORMAL / UNUSUAL / EXTREME**, supported by active NWS alerts and by the impact history of the same county.

Scope. State of Texas; 10 first-order ASOS stations (KDFW, KIAH, KAUS, KSAT, KELP, KLBB, KAMA, KCRP, KMAF, KBRO) that are published both by GHCN-Daily and by the NWS API; all Storm Events of Texas counties and forecast zones; all NWS alerts with `area=TX`. Historical window 2010-01-01 → present.

Explicit exclusions. No forecasting or prediction of hazards; no numerical weather models or radar/satellite imagery; no states other than Texas; no marine zones beyond what NWS includes in `area=TX`; no damage estimation; no individual-level or personal data.

Constraints. Free and open sources only; US-only services (hence US scope); NWS requires a descriptive User-Agent; the observations endpoint returns at most 500 features; GHCN-Daily has a ~3-day publication delay and Storm Events several months; local execution with Docker Compose; all documentation in English.

2. Hazard Definition

Phenomenon. Extreme weather is weather at the tails of the local climatological distribution — values that are rare for *that place and that time of year*. Team 5 monitors seven hazard components:

Component	How it is measured	Severity measure and unit	Sources
Extreme heat	Daily maximum temperature; instantaneous temperature and heat index	TMAX, heat index (°C)	GHCN-Daily, NWS obs, Storm Events (Heat / Excessive Heat), NWS heat alerts
Extreme cold	Daily minimum temperature; wind chill	TMIN, wind chill (°C)	GHCN-Daily, NWS obs, Storm Events (Cold/Wind Chill, Extreme Cold, Frost/Freeze), freeze alerts
Heavy precipitation / flash flood	24-h precipitation accumulation; 1-h rate	PRCP (mm/day), precipitation 1 h (mm)	GHCN-Daily, NWS obs, Storm Events (Heavy Rain, Flash Flood), flood alerts
Severe wind	Fastest 5-second gust; measured/estimated gust in reports	WSF5, gust (m/s); Storm Events knots → m/s	GHCN-Daily, NWS obs, Storm Events (Thunderstorm/High/Strong Wind), wind alerts
Hail	Spotter/observer hail size	Diameter (in → mm)	Storm Events, Severe Thunderstorm Warnings
Tornado	Damage survey	EF scale (0–5)	Storm Events, Tornado Watch/Warning
Winter storm	Daily snowfall and snow depth	SNOW, SNWD (mm)	GHCN-Daily, Storm Events (Winter Storm, Heavy Snow, Ice Storm), winter alerts

Alerts carry an ordinal severity (CAP: Minor 1, Moderate 2, Severe 3, Extreme 4). Storm Events without a physical magnitude receive an impact level 0–3 derived from casualties and damage (§8).

Why it matters. Heat is the leading weather-related cause of death in the US, freezes stress the Texas power grid (February 2021), and Texas is among the states with the most flood fatalities, most of them in flash floods. Knowing *early* that a measurement is unusual for the station and season helps emergency managers decide on cooling/warming centres, public messaging and resource staging.

Geographic scope and justification. Texas is mandatory-compatible with all four sources (Storm Events and NWS are US services; GHCN-Daily's densest coverage is the US). Ten first-order stations give a manageable volume (≈62,000 station-days for 2010–2026) while covering the climatic diversity of the state: humid Gulf coast (KIAH, KCRP, KBRO), central Texas flash-flood alley (KAUS, KSAT), north Texas (KDFW), Panhandle and South Plains (KAMA, KLBB), Permian Basin (KMAF) and the Chihuahuan desert (KELP, Mountain Time).

Historical window. 2010-01-01 → present (≥ 16 years, above the five required). Sixteen years give ≈ 500 values per station for a ±15-day seasonal window, enough to estimate the 99th percentile; going further back would add station relocations and longer-term climate trends that bias a "current climate" baseline, without improving the tail estimate much.

Current situation. (a) NWS alerts active *now* for Texas; (b) station observations of the *last 24 hours* (rolling max/min temperature, 24-h precipitation, max gust) compared with the baseline; (c) the *last 7 days* of alerts and observations for context. Storm Events and GHCN-Daily are too delayed to define the current situation; they provide the baseline and the impact history.

3. Data Sources and Validation Evidence

All four sources were queried for real on 2026-11-09 with `workshop_1/scripts/fetch_samples.py`. Every response is stored unchanged with its HTTP headers in `workshop_1/samples/` (and `data/raw/` for the 12 MB yearly file).

	H1 NOAA Storm Events	H2 GHCN-Daily (NCEI ADS)	I1 NWS alerts	I2 NWS observations
Owner	NOAA NCEI	NOAA NCEI	NOAA National Weather Service	NOAA National Weather Service
URL	ncei.noaa.gov/pub/data/swdi/stormevents/csvfiles/	ncei.noaa.gov/access/services/data/v1	api.weather.gov/alerts	api.weather.gov/stations/{id}/observations
Access	HTTPS bulk download, one file per year	REST GET with query parameters	REST GET, cursor pagination	REST GET per station
Authentication	None	None	None; User-Agent required	None; User-Agent required
Rate limits	None published; download once	None published; keep requests small	Not published; 403/429 on abuse	Not published; 500 features per response
License	US Government, public domain; cite NCEI	Public domain; cite Menne et al. (2012)	Public domain	Public domain
Format	CSV.gz, 51 columns	JSON (also CSV), 22 keys	GeoJSON + CAP 1.2 fields	GeoJSON, WMO unit codes
Key fields	EVENT_ID, EPISODE_ID, EVENT_TYPE, BEGIN_*, CZ_TYPE/FIPS/NAME, CZ_TIMEZONE, MAGNITUDE(_TYPE), damages, BEGIN_LAT/LON	STATION, DATE, TMAX, TMIN, PRCP, SNOW, SNWD, AWND, WSF2, WSF5, *_ATTRIBUTES, LATITUDE, LONGITUDE	id, sent, effective, onset, expires, messageType, event, severity, certainty, urgency, geocode.UGC, references, geometry	stationId, timestamp, temperature, windSpeed, windGust, precipitationLastHour, heatIndex, windChill (value, unitCode, qualityControl)
Time coverage	1950 → current year (several months of delay)	Station records since 1940s; last day 2026-11-06 (3-day delay)	Last ~7 days (oldest sent 2026-11-02)	Last ~4 days per request (500 cap)
Update policy	Monthly; whole years recompiled (`_cYYYYMMDD`)	Daily; values re-sourced/QC'd later	Continuous (minutes)	Every 5–60 min per station

3.1 H1 — Storm Events (header + one Texas row, unmodified)

```
GET https://www.ncei.noaa.gov/pub/data/swdi/stormevents/csvfiles/StormEvents_details-ftp_v1.0_d2025_c20260819.csv.gz
{"Date": "Mon, 09 Nov 2026 23:22:23 GMT", "Last-Modified": "Wed, 19 Aug 2026 22:22:37 GMT", "ETag": "\"c008f9-6596dd737e99d\"", "Content-Length": "1258520
9", "Content-Type": "application/gzip"}

BEGIN_YEARMONTH,BEGIN_DAY,BEGIN_TIME,END_YEARMONTH,END_DAY,END_TIME,EPISODE_ID,EVENT_ID,STATE,STATE_FIPS,YEAR,MONTH_NAME,EVENT_TYPE,CZ_TYPE,CZ_FIPS,CZ_NAME,
WFO,BEGIN_DATE_TIME,CZ_TIMEZONE,END_DATE_TIME,INJURIES_DIRECT,INJURIES_INDIRECT,DEATHS_DIRECT,DEATHS_INDIRECT,DAMAGE_PROPERTY,DAMAGE_CROPS,SOURCE,MAGNITUDE,
MAGNITUDE_TYPE,FLOOD_CAUSE,CATEGORY,TOR_F_SCALE,TOR_LENGTH,TOR_WIDTH,TOR_OTHER_WFO,TOR_O …
202503,4,1008,202503,4,1008,199707,1235565,"TEXAS",48,2025,"March","Thunderstorm Wind","C",201,"HARRIS","HGX","04-MAR-25 10:08:00","CST-6","04-MAR-25 10:08:
00","0","0","0","0","0.00K","0.00K","Mesonet","69.00","MG",,,,,,,,,,".69","S","WHITE OAK ACRES",".69","S","WHITE OAK ACRES","29.81","-95.45","29.81","-95.4
5","Thunderstorms along a cold front produced damaging winds, resulting in downed trees/b …
```

3.2 H2 — GHCN-Daily (first record of the JSON response)

```
GET https://www.ncei.noaa.gov/access/services/data/v1?dataset=daily-summaries&stations=USW00003927%2CUSW00012960%2CUSW00013904%2CUSW00012921%2CUSW00023044%2CUSW00023042%2CUSW00023047%2CUSW00012924%2CUSW00023023%2CUSW00012919&startDate=2025-01-01&endDate=2026-11-09…
{"Date": "Mon, 09 Nov 2026 23:22:25 GMT", "Content-Type": "application/json;charset=utf-8"}

{
 "WSF2": "6.7",
 "WSF5_ATTRIBUTES": ",,W",
 "STATION": "USW00003927",
 "WSF5": "8.5",
 "SNOW": "0.0",
 "LONGITUDE": "-97.02196",
 "PRCP_ATTRIBUTES": ",,W",
 "ELEVATION": "165.7",
 "SNOW_ATTRIBUTES": ",,1",
 "PRCP": "0.0",
 "AWND_ATTRIBUTES": ",,W",
 "SNWD_ATTRIBUTES": ",,1",
 "SNWD": "0.0",
 "NAME": "DAL FTW WSCMO AIRPORT, TX US",
 "DATE": "2025-01-01",
 "AWND": "2.9",
 "TMAX": "10.0",
 "TMAX_ATTRIBUTES": ",,W",
 "TMIN": "2.2",
 "WSF2_ATTRIBUTES": ",,W",
 "LATITUDE": "32.89744",
 "TMIN_ATTRIBUTES": ",,W"
}
```

3.3 I1 — NWS alerts (one feature, selected properties)

```
GET https://api.weather.gov/alerts?area=TX&limit=500
{"Content-Type": "application/geo+json", "ETag": "W/\"8bd1b0d5a58fdfb466281d18e6400703\"", "X-Correlation-ID": "30aa1b1", "Cache-Control": "public, max-age=5, s-maxage=5", "Expires": "Mon, 09 Nov 2026 23:22:36 GMT", "Date": "Mon, 09 Nov 2026 23:22:31 GMT"}

{
 "geometry": "(Polygon, 7 vertices)",
 "properties": {
  "id": "urn:oid:2.49.0.1.840.0.f221cdce3f47d63a72f8779b4f7cd8516dd7d626.001.1",
  "sent": "2026-11-09T17:02:00-06:00",
  "effective": "2026-11-09T17:02:00-06:00",
  "onset": "2026-11-09T23:06:00-06:00",
  "expires": "2026-11-10T05:15:00-06:00",
  "status": "Actual",
  "messageType": "Update",
  "severity": "Severe",
  "certainty": "Likely",
  "urgency": "Expected",
  "event": "Flood Warning",
  "areaDesc": "Llano, TX",
  "geocode": {
   "SAME": [
    "048299"
   ],
   "UGC": [
    "TXC299"
   ]
  },
  "references": [
   {
    "@id": "https://api.weather.gov/alerts/urn:oid:2.49.0.1.840.0.7a61b93708ec2a3e7684e863eec56c1de7d96ddd.001.1",
    "identifier": "urn:oid:2.49.0.1.840.0.7a61b93708ec2a3e7684e863eec56c1de7d96ddd.001.1",
    "sender": "w-nws.webmaster@noaa.gov",
    "sent": "2026-11-09T13:13:00-06:00"
   }
  ]
 },
 "pagination": {
  "next": "https://api.weather.gov/alerts?area%5B0%5D=TX&limit=500&cursor=eyJ0IjoxNzkzNjYxODQwLCJpIjoidXJuOm9pZDoyLjQ5LjAuMS44NDAuMC5hZTI2OWNjODUyNDZjMzY5YzE0ZmY5NTRiOWEwYzRjNTJlYmNiMGRhLjAwMS4xIn0%3D"
 }
}
```

3.4 I2 — NWS observations (first feature for KDFW, selected properties)

```
GET https://api.weather.gov/stations/KDFW/observations?start=2026-11-02T23%3A22%3A27Z
{"Content-Type": "application/geo+json", "X-Correlation-ID": "27a4aa1a", "Content-Length": "39613", "Cache-Control": "public, max-age=300, s-maxage=120", "E
xpires": "Mon, 09 Nov 2026 23:27:32 GMT", "Date": "Mon, 09 Nov 2026 23:22:32 GMT"}

{
 "geometry": {
  "type": "Point",
  "coordinates": [
   -97.03,
   32.9
  ]
 },
 "properties": {
  "stationId": "KDFW",
  "timestamp": "2026-11-09T22:45:00+00:00",
  "textDescription": "Cloudy",
  "temperature": {
   "unitCode": "wmoUnit:degC",
   "value": 21,
   "qualityControl": "V"
  },
  "windSpeed": {
   "unitCode": "wmoUnit:km_h-1",
   "value": 9.252,
   "qualityControl": "V"
  },
  "windGust": {
   "unitCode": "wmoUnit:km_h-1",
   "value": null,
   "qualityControl": "Z"
  },
  "precipitationLastHour": "(key absent)",
  "heatIndex": {
   "unitCode": "wmoUnit:degC",
   "value": null,
   "qualityControl": "V"
  }
 }
}
```

4. Source Profiling

The notebook `notebooks/01_source_profiling.ipynb` profiles every source (≥ 200 records each): null percentage per field, value ranges, duplicated identifiers, timestamp format and time zone, coordinate validity and unit consistency, and applies the draft canonical mapping. Summary after mapping:

source	raw_records	canonical_events	duplicate_ids	first_utc	last_utc	no_coordinates_pct
ghcn_daily	6,747 station-days	53802	0	2025-01-01 06:00	2026-11-06 07:00	0.0
noaa_storm_events	5,641 rows	5641	0	2025-01-01 06:00	2025-12-31 10:00	36.7
nws_alerts	224 alerts	224	0	2026-11-02 23:24	2026-11-09 23:11	26.3
nws_observations	5,000 observations	11803	0	2026-11-05 16:30	2026-11-09 23:00	0.0

Main observations: no duplicated native identifiers in any source; Storm Events uses local standard time per county/zone (`CST-6` for 98% of rows, `MST-7` for El Paso and Hudspeth counties/zones); 36.7% of Storm Events rows (all zone-based, `CZ_TYPE='Z'`) have no coordinates; GHCN-Daily completeness is 99.9–100% for the ten stations with no TMIN>TMAX inconsistencies; NWS alerts use local offsets (`-06:00`); NWS observations are already UTC, but 96% of wind-gust values and 97% of hourly precipitation values are null.

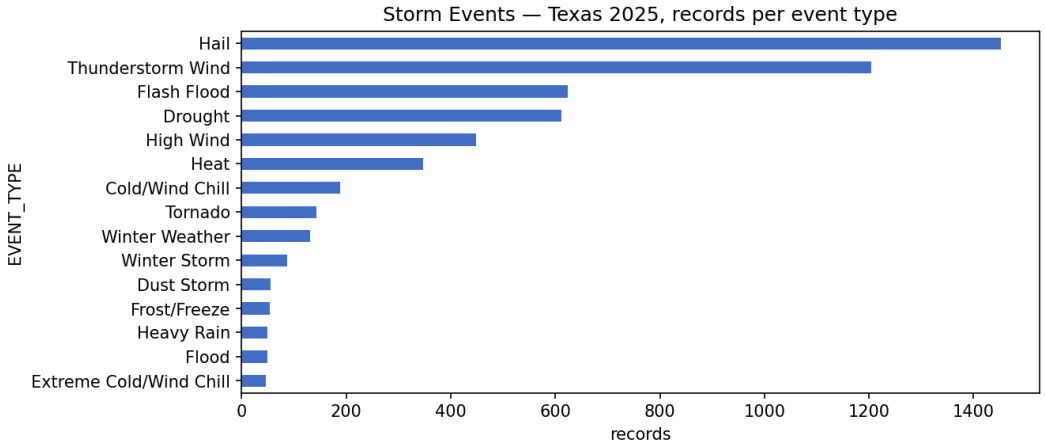


Figure 1. Storm Events, Texas 2025 — records per event type (top 15).

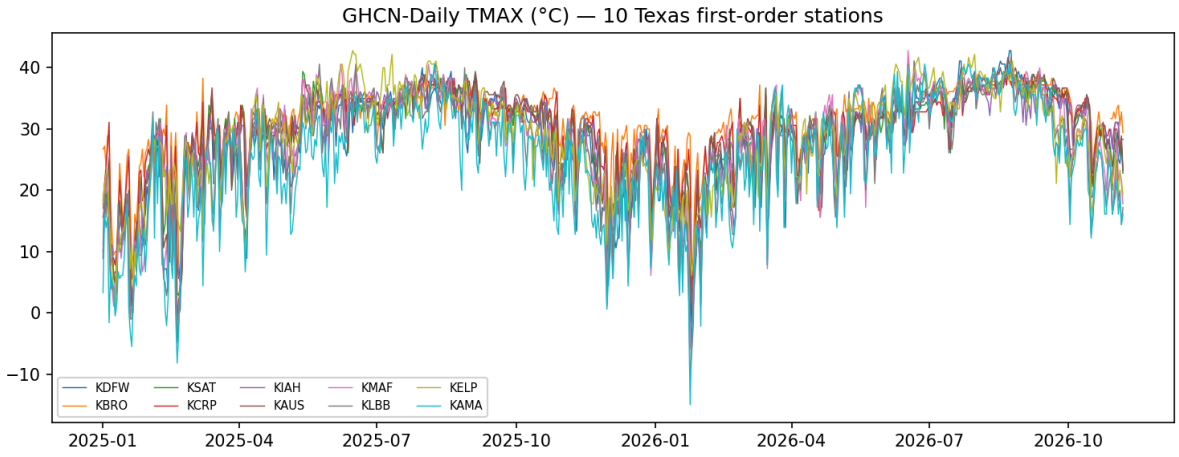


Figure 2. GHCN-Daily TMAX for the ten stations, 2025-01-01 → 2026-11-06.

Volume estimate and partitioning. Storm Events 2010–2026: 17 yearly files of 8.6–15.0 MB gz (≈ 186 MB US-wide, ≈ 96,000 Texas rows at ≈ 5,600 per year) → partition by year. GHCN-Daily: ≈ 62,000 station-days × 8 elements (≈ 500,000 measurements, ≈ 28 MB JSON) → 170 station-year requests. NWS alerts ≈ 11,600 messages/year for Texas. NWS observations ≈ 450,000 observations/year for 10 stations (~5-min reporting) → polled every 30 min per station.

5. Data-Quality Findings

Thirteen concrete problems were found (five required); each has a real example record from the samples. The full table is saved as `workshop_1/data_quality_issues.csv`.

source	problem	extent	example record
H1 Storm Events	No coordinates for zone/county-based events	36.7% of rows	EVENT_ID 1230373: Coastal Flood, CZ_TYPE=Z, CZ_NAME=COASTAL GALVESTON, BEGIN_LAT=NaN
H1 Storm Events	Local standard time + 2-digit year in BEGIN_DATE_TIME	all rows	EVENT_ID 1230373: BEGIN_DATE_TIME='09-JAN-25 20:00:00', CZ_TIMEZONE='CST-6'
H1 Storm Events	Damage stored as text with K/M/B suffix; blanks mean 'unknown', not 0	15.7% blank	EVENT_ID 1256818: DAMAGE_PROPERTY='10.00K'
H1 Storm Events	MAGNITUDE has different units per event type (hail inches, wind knots; EG=estimated vs MG=measured)	Hail/Wind rows	EVENT_ID 1230425 Hail MAGNITUDE=1.00 (in); EVENT_ID 1244626 TSTM Wind MAGNITUDE=64.00 EG (kt)
H1 Storm Events	CZ_FIPS of forecast zones (Z) is a zone number, not a county FIPS; names vary ('INLAND HARRIS')	36.7% of rows	EVENT_ID 1230373: CZ_TYPE=Z, CZ_FIPS=338, CZ_NAME=COASTAL GALVESTON
H1 Storm Events	Filtering by the text 'TEXAS' also matches Texas County (MO, OK) and other states' fields; filter must use STATE_FIPS=48	66 wrong rows (MISSOURI: 34, OKLAHOMA: 31, MARYLAND: 1)	EVENT_ID 1251602: STATE=MISSOURI, CZ_NAME=TEXAS, TOR_OTHER_CZ_NAME=nan
H2 GHCN-Daily	Implausible wind gust value not flagged by QC	1+ rows	USW00023042 2025-06-17: WSF5=76.0 m/s (~148 kt), attributes=',,W'

source	problem	extent	example record
H2 GHCN-Daily	Missing elements are absent keys (not null); numbers arrive as strings	WSF5 absent in 0.5%	USW00003927 2025-03-09: keys without WSF5; TMAX='14.4' (str)
H2 GHCN-Daily	Latency: most recent days not yet published	last DATE 2026-11-06	query endDate = download date; last available = 2026-11-06
I1 NWS alerts	Alerts without polygon geometry (only UGC zone codes)	26.3%	d7bcdd5771c0a3.002.1: event=Coastal Flood Advisory, geometry=None, UGC=['TXZ245', 'TXZ342', 'TXZ343']
I1 NWS alerts	Timestamps carry local offsets (-05:00/-06:00/-07:00), not UTC	all rows	sent='2026-11-09T17:11:00-06:00'
I2 NWS observations	Silent 500-feature cap truncates the requested 7-day window	every station (min 91.3 h of 168 h)	KDFW: 500 records from 2026-11-05 23:00:00+00:00 to 2026-11-09 22:45:00+00:00
I2 NWS observations	Nulls encoded as value=None with qualityControl='Z'; some keys absent	windGust null 96%	KAMA 2026-11-09T22:45:00+00:00: windGust={'value': None, 'qualityControl': 'Z'}

> **Most relevant for the design:** (1) silent truncation of NWS observations at 500 features — a naive "last 7 days" request loses 40% of the window, so the incremental poll must be frequent and detect full pages; (2) a GHCN-Daily WSF5 of 76.0 m/s (~148 kt) at Lubbock on 2025-06-17 with a blank QC flag, while Storm Events reports a 50-kt measured gust at the same county that day: the validation layer needs physical-plausibility ranges in addition to source QC flags; (3) zone-based Storm Events have neither coordinates nor county FIPS, so matching requires a zone lookup table, never name matching (`HARRIS` also matches `HARRISON`).

6. Incremental Behaviour — Two Polls at Least One Hour Apart

The incremental sources were polled twice with the same requests; identifiers, content hashes and HTTP caching headers were compared (notebook §8).

Measure	Poll 1	Poll 2	What changed
Poll time (UTC)	20261109_232218	20261110_002450	—
Active alerts TX	10	8	0 new, 2 no longer active, 8 unchanged ids
Recent alerts TX (/alerts)	224	202	0 new ids; 22 left the window; 0 same id with changed content
Max alert 'sent'	2026-11-09T17:11:00-06:00	2026-11-09T17:11:00-06:00	watermark candidate moves forward
Observations (10 stations)	5000	5000	38 new (station, timestamp); 0 existing keys with revised values
Max obs timestamp KDFW	2026-11-09T22:45:00+00:00	2026-11-09T23:55:00+00:00	per-station watermark
Alerts ETag / Last-Modified	W/"0aba9408137237a9c7a633c9848b5192" / —	W/"b2477b055f8fac217e56fa4dff845fe6" / —	
Obs KDFW Expires / Last-Modified	Mon, 09 Nov 2026 23:27:32 GMT / —	Tue, 10 Nov 2026 00:29:56 GMT / —	

How new or modified records can be detected. NWS alerts: every message has a stable, globally unique `id`; an alert never changes in place — it evolves through new `Update`/`Cancel` messages that list superseded ids in `references`, and leaves `/alerts/active` when it expires. Detection therefore uses *new ids since the watermark on `sent`*, plus the reference chains. NWS observations: the key is `(stationId, timestamp)`; the API accepts a `start` parameter, which is the natural "updated since" filter. Caching headers are only partly useful: `/alerts/active` returns a weak `ETag` with `max-age=5`, which allows a conditional request (`If-None-Match`) to skip an unchanged active set, but it says nothing about *which* records changed; the observations endpoint and the NCEI Access Data Service return neither ETag nor Last-Modified. Record-level content hashes are therefore the change signal.

Revisions and deletions. Storm Events recompiles whole years long after publication (the 2012 file was recompiled on 2026-10-26); GHCN-Daily station-days are re-published from another source stream about a year later (source flag `W` before Aug-2025, `1` afterwards); NWS observations can receive late QC; alerts are never edited but disappear from the active set. Consequence for the watermark: it is never a strict ">" filter — each source re-reads an overlap window (2 h alerts, 3 h observations, 45 days + monthly 13-month re-read GHCN, whole year Storm Events) and the content hash decides between insert, update or no-op.

7. Cross-Source Overlap

Rule used for candidates (refined in Workshop 3): Storm Events record in the county of a station (exact `CZ_FIPS` for county rows; word-boundary name match for zone rows) and on the same local calendar day as the GHCN-Daily record, with a compatible variable. The rule produced 471 candidate pairs for 2025; the strongest pair per event type is listed:

Storm Events record	Zone/county	Begin (local)	SE magnitude	GHCN record	GHCN value
SE 1272279 — Flash Flood	C:BEXAR	12-JUN-25 01:15:00 CST-6	no magnitude	USW00012921 2025-06-12	PRCP = 155.2 mm

Storm Events record	Zone/county	Begin (local)	SE magnitude	GHCN record	GHCN value
SE 1246058 — Winter Storm	Z:POTTER	08-MAR-25 19:00:00 CST-6	no magnitude	USW00023047 2025-03-08	SNOW = 102.0 mm
SE 1271694 — Thunderstorm Wind	C:LUBBOCK	17-JUN-25 18:19:00 CST-6	50.00 kt MG = 25.7 m/s	USW00023042 2025-06-17	WSF5 = 76.0 m/s
SE 1294685 — Heavy Rain	C:EL PASO	19-SEP-25 20:13:00 MST-7	no magnitude	USW00023044 2025-09-19	PRCP = 49.5 mm
SE 1246090 — High Wind	Z:POTTER	14-MAR-25 05:46:00 CST-6	72.00 kt MG = 37.0 m/s	USW00023047 2025-03-14	WSF5 = 37.1 m/s
SE 1287962 — Heat	Z:TARRANT	24-JUL-25 11:00:00 CST-6	no magnitude	USW00003927 2025-07-24	TMAX = 36.7 °C
SE 1233513 — Extreme Cold/Wind Chill	Z:TARRANT	18-FEB-25 22:00:00 CST-6	no magnitude	USW00003927 2025-02-18	TMIN = -7.1 °C
SE 1234898 — Cold/Wind Chill	Z:TARRANT	06-JAN-25 00:00:00 CST-6	no magnitude	USW00003927 2025-01-06	TMIN = -6.0 °C
SE 1235712 — Frost/Freeze	Z:COASTAL NUECES COUNTY	21-JAN-25 00:00:00 CST-6	no magnitude	USW00012924 2025-01-21	TMIN = -2.7 °C

What differs. *Time:* Storm Events gives a local begin time to the minute, GHCN a local-standard *day*; matching must happen on the local day, not the UTC day. *Position:* zone/county polygons versus a station point (zone rows have no coordinates). *Severity:* knots versus m/s and estimated (EG) versus measured (MG) gusts — the High Wind pair at Potter County (72 kt MG = 37.0 m/s vs WSF5 = 37.1 m/s) agrees within 0.1 m/s, whereas the Lubbock Thunderstorm Wind pair (50 kt vs 76.0 m/s) exposes the GHCN outlier of §5. Heat/cold/flood reports have no magnitude, so the station value is the only physical measurement. *Identifier:* no shared key at all.

Second overlap — NWS observations ↔ GHCN-Daily. The same ASOS sensors feed both services. For complete local days, the maximum of the NWS observations matches GHCN TMAX with a mean absolute difference of 0.11 °C over 9 complete station-days when days are cut at local standard midnight, versus 0.22 °C when the UTC day is used (notebook §7) — confirming both the station pairing and the local-standard-day rule of §8; incomplete days at the start of the 500-record window must be excluded, another consequence of the truncation. **Third overlap — alert chains:** 105 of 224 alerts reference earlier messages of the same event (`Update`/`Cancel`), so even within one source several records describe one real-world event.

8. Canonical Mapping and Domain Tables

The full field-by-field mapping is in `workshop_1/mapping.md` and is implemented as a draft in `src/normalization/canonical.py` (17 pytest tests on recorded fixtures pass).

Canonical	H1 Storm Events	H2 GHCN-Daily	I1 NWS alerts	I2 NWS observations
event_id	noaa_storm_events:EVENT_ID	ghcn_daily:STATION:DATE:ELEMENT	nws_alerts:properties.id	nws_observations:stationId:timestamp:variable
source	constant per source (FK to `source`)			
event_type	EVENT_TYPE → snake_case	element → daily_max_temperature, daily_precipitation…	event → snake_case	variable → air_temperature, wind_gust…
timestamp (UTC)	BEGIN_YEARMONTH+DAY+TIME – offset(CZ_TIMEZONE)	DATE local-standard midnight → UTC (06:00Z / 07:00Z)	effective (ISO offset → UTC)	timestamp (already UTC)
latitude, longitude	BEGIN_LAT/LON; zone rows → zone centroid	LATITUDE/LONGITUDE	polygon centroid; no geometry → UGC zone centroid	geometry [lon, lat] (station table preferred)
severity + unit	hail mm; wind m/s (EG/MG); EF; else impact 0–3	value °C / mm / m/s	CAP severity 0–4	value converted by unitCode
raw_reference	EVENT_ID + file compilation; SHA-256 hash	STATION/DATE/ELEMENT; hash of value+flags	@id URL; hash	@id URL; hash of value+QC

Rules. UTC everywhere; WGS84 decimal degrees, validated against the Texas bounding box (+0.5°); °C, mm, m/s; unit conversion driven by the unit declared in each record (an unknown unit is quarantined); timestamps built from the 4-digit-year fields, never from the 2-digit `BEGIN_DATE_TIME`; blank damage = unknown (NULL), not zero.

Domain tables. `weather_station` (static pairing GHCN ↔ NWS, LST offset), `weather_measurement` (element, period P1D/PT1H, value, raw value/unit, GHCN flags, NWS QC code), `storm_report` (episode, CZ type/FIPS/name, magnitude and type, casualties, parsed damages, tornado attributes, narrative, file compilation), `weather_alert` (CAP id, message type, severity/certainty/urgency, sent/onset/expires/ends, VTEC, polygon), `alert_zone` and `alert_reference` (bridges), `zone` (UGC county/forecast-zone polygons), and in Workshop 4 `station_baseline` (percentiles per station, element and seasonal window). Draft DDL: `sql/001_schema_draft.sql`.

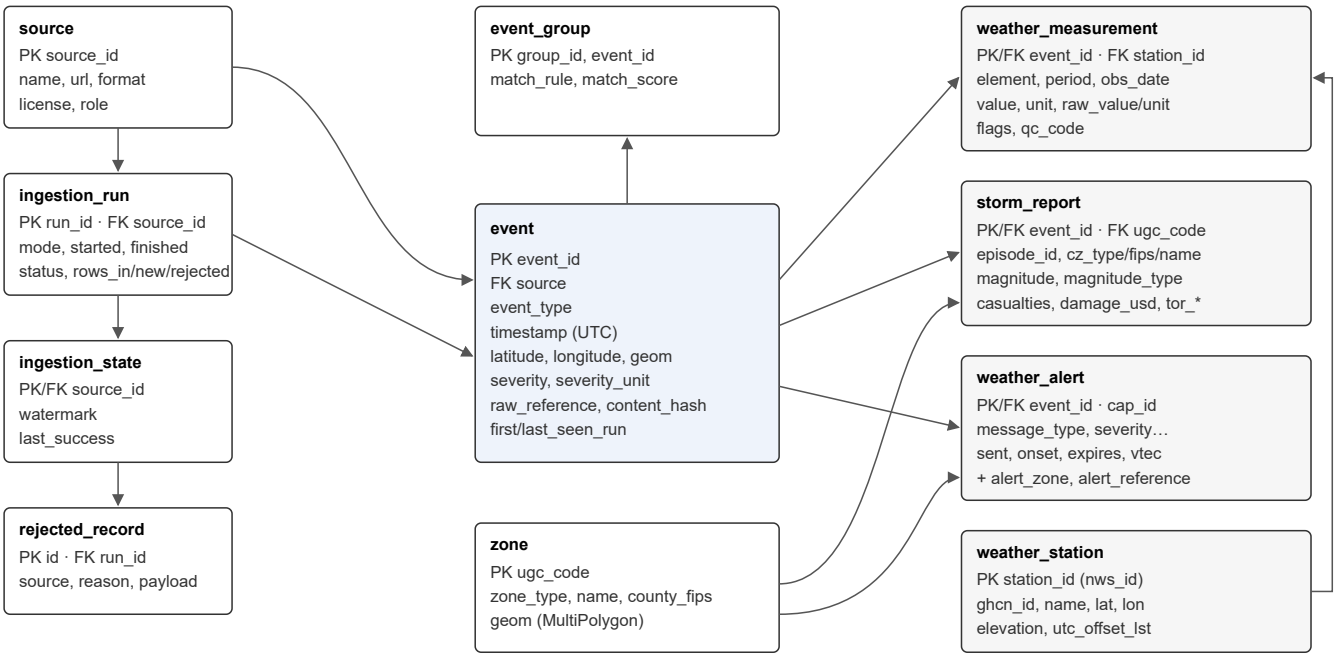


Figure 3. Draft ER diagram: common contract (left/centre) and Team 5 domain tables (right). Arrows point from parent to child.

9. Ingestion Contract (Draft)

	H1 Storm Events	H2 GHCN-Daily	I1 NWS alerts	I2 NWS observations
Historical	17 yearly CSV.gz (2010→), filter TEXAS; partition = year	10 stations × 17 years = 170 JSON requests; partition = station-year	— (API keeps ~7 days)	— (500-feature cap)
Incremental	Monthly listing check; reload year when `_cYYYYMMDD` changes	Daily `[wm − 45 d, today]` per station; monthly 13-month re-read	Every 15 min `?area=TX&start=wm−2h`, follow `pagination.next`; refresh `/active`	Every 30 min per station `?start=wm−3h`; split window if 500 rows
Watermark	{year: compilation date}	max DATE per station	max `sent`	max `timestamp` per station
Overlap	whole year	45 days	2 h	3 h
Duplicate key	EVENT_ID	(STATION, DATE, ELEMENT)	properties.id	(stationId, timestamp, variable)
Change detection	compilation suffix + hash; missing rows → withdrawn flag	hash of value + attributes	new ids + reference chains	hash of value + QC code

Correctness if a run fails halfway. (1) The raw response is written to the landing zone (`source_YYYYMMDD_HHMMSS.ext`) before parsing, so it can be replayed. (2) For each source, canonical rows, domain rows, quarantined rows and the new watermark are committed in a single transaction; if the process dies, the transaction rolls back and the watermark keeps its last successful value. (3) The next run re-requests the same interval plus the overlap; deterministic `event_id`s and `INSERT … ON CONFLICT DO UPDATE WHERE content_hash differs` make the repetition harmless. (4) Historical partitions are logged when committed, so a restart resumes from the first unfinished year/station-year. (5) The failed `ingestion_run` keeps `status='failed'` and the error, visible in `--mode status` and in the dashboard's data-health panel. Details: `docs/ingestion_contract.md`.

10. Functional User

Role. Emergency-management analyst in a Texas county Office of Emergency Management (or a regional office of the Texas Division of Emergency Management).

Profile. Not a meteorologist; comfortable with dashboards and maps; works shifts; reads NWS products but has no time to cross-check several websites; must brief decision makers twice a day and justify recommendations with facts.

Decisions supported. Whether to open cooling or warming centres; whether to raise the county's readiness level or pre-position resources (flood barricades, crews); what to say in public messaging ("hottest night for this time of year in 15 years"); whether data are fresh enough to trust before a briefing.

Why the dashboard is useful. It puts in one screen what is happening *now* (alerts + latest observations), whether it is *unusual for this place and season* (percentile-based level), what happened in the past in the same county (Storm Events impacts) and how *fresh and complete* the data are — without claiming to predict anything.

11. User Stories (MoSCoW)

#	Priority	Theme	User story
US1	**Must**	Monitoring	As an emergency-management analyst, I want to see the NWS alerts active now for Texas on a map with their severity so that I know which counties are under official warnings.
US2	**Must**	Monitoring	As an analyst, I want a list of what is new since the previous ingestion (new, updated and cancelled alerts, new observations) so that I do not miss changes between briefings.
US3	**Must**	Analysis	As an analyst, I want each station's latest 24-h max/min temperature, precipitation and gust classified as NORMAL / UNUSUAL / EXTREME against the same station and season so that I can tell whether conditions are truly unusual.
US4	**Must**	Trust	As an analyst, I want to see for each source the last successful run, the watermark age and the number of rejected records so that I know how fresh and complete the data are before briefing.
US5	**Should**	Analysis	As an analyst, I want a time series of the last 30 days for a station drawn over its historical percentile band so that I can explain trends to decision makers.
US6	**Should**	Analysis	As an analyst, I want to see past Storm Events (type, date, damage, casualties) in the selected county for situations of the same type so that I can anticipate likely impacts.
US7	**Should**	Trust	As an analyst, I want to see which records were grouped as the same event across sources (alert, observation, storm report) so that I can trust that counts are not duplicated.
US8	**Could**	Analysis	As an analyst, I want to change the percentile thresholds of the indicator and see how many stations change level so that I understand its sensitivity.
US9	**Could**	Monitoring	As an analyst, I want to export the current classification table as CSV so that I can attach it to my briefing.
US10	**Won't (this semester)**	—	As an analyst, I want a forecast of tomorrow's hazard level. Excluded: the system is a monitoring aid, not a prediction tool.

12. Indicator Proposal — Extreme Weather Level (NORMAL / UNUSUAL / EXTREME)

Inputs. (a) Baseline: GHCN-Daily 2010 → last complete year for each station and element (TMAX, TMIN, PRCP, WSF5, SNOW), using a ±15-day window around each day of the year (≈ 31 days × 16 years ≈ 500 values) → percentiles p1, p5, p10, p90, p95, p99 and record values. (b) Current values: from NWS observations, rolling 24-h maximum and minimum temperature, 24-h precipitation sum and maximum gust; from GHCN, the latest published day. (c) Active NWS alerts whose zones contain the station.

Rule (parameterised; defaults shown). For "high-tail" elements (TMAX, PRCP, WSF5, SNOW): NORMAL if value < p90; UNUSUAL if p90 ≤ value < p99; EXTREME if value ≥ p99 or above the seasonal record. For TMIN the mirror rule applies (≤ p10 UNUSUAL, ≤ p1 EXTREME). Precipitation percentiles are computed on wet days only (PRCP ≥ 1 mm) to avoid a distribution dominated by zeros. *Alert escalation:* if an active alert of the matching type (e.g., Extreme Heat Warning, Freeze/Hard Freeze Warning, Flash Flood Warning, High Wind Warning) covers the station, the level is raised one step (capped at EXTREME) and marked "official alert". Output per station × component: level, percentile rank, alert flag; per region: the worst component level and the number of stations at each level.

Validation against historical events. Back-test the rule on 2010–2025 daily GHCN values with a leave-one-year-out baseline, and compare with Storm Events of the matching type in the station's county/zone on the same local day: *hit rate* = share of Storm Events days of that type flagged UNUSUAL or EXTREME; *false-alarm share* = share of EXTREME days without any matching report; plus case studies (February 2021 winter storm, summer 2023 heat, 2025 central Texas floods). A sensitivity table over p90/p95/p97.5 and p99/p99.5 thresholds will be reported (Workshop 4).

Limitations. Percentiles describe rarity, not impact; ten stations do not represent every county; Storm Events reports are biased toward populated areas; GHCN-Daily and the NWS day definitions differ (local standard day vs UTC), which is handled explicitly; the indicator is a monitoring aid and **does not predict** hazards.

13. Risks and Mitigation

#	Risk	Likelihood / impact	Mitigation
R1	Source outage (NWS API 5xx, NCEI maintenance)	Medium / High	Retries with exponential back-off and timeouts; watermark advances only after commit; raw cache allows re-processing; data-health panel shows watermark age.
R2	Rate limiting or blocking (403 without User-Agent, 429)	Medium / High	Descriptive User-Agent from `.env`; polling at 15/30 min; ≥ 0.5 s between requests; cache every download; never re-download the archive.
R3	Silent truncation (500-feature cap on observations)	High (observed) / Medium	Frequent polls with small windows; if a response has 500 rows, split the window and re-request; count check per run.
R4	Schema change (new/renamed fields, unit codes, new CSV columns)	Medium / High	Explicit expected-schema checks on read; unit conversion by declared unit; unknown units/fields quarantined; recorded fixtures in tests; source-status log and notice to the professor.
R5	Revisions and withdrawals (Storm Events recompilations, GHCN re-publication, late QC)	High (observed) / Medium	Overlap windows, content hashes, compilation-suffix watermark, soft "withdrawn" flag instead of deletes.

R6	Ambiguous identifiers/locations (zone vs county FIPS, name variants, alerts without polygons)	High (observed) / High	Static `zone` table from NWS `/zones` and Census counties; match on UGC/FIPS codes, never on names; centroid with `location_method` flag.
R7	Data volume (≈ 186 MB Storm Events, ≈ 450 k observations/year)	Medium / Medium	Filter Texas at read time; partitioned loads; indexes on (event_type, timestamp) and GiST; daily aggregate views for the dashboard.
R8	Latency (GHCN 3 days, Storm Events months) and off-season semester (few heat events in Nov–Dec)	High / Medium	"Current" is defined with NWS sources; GHCN/Storm Events used for baseline; cold, wind and rain components remain active in winter; recorded fixtures from past extreme days for demos.
R9	License / attribution non-compliance	Low / Medium	All sources are public domain; attribution in `config/sources.yaml`, README and report; GHCN DOI cited.
R10	Misinterpretation of the indicator as a forecast	Medium / High	Labels "monitoring aid", visible limitations panel, link to official NWS products.

14. Architecture Overview

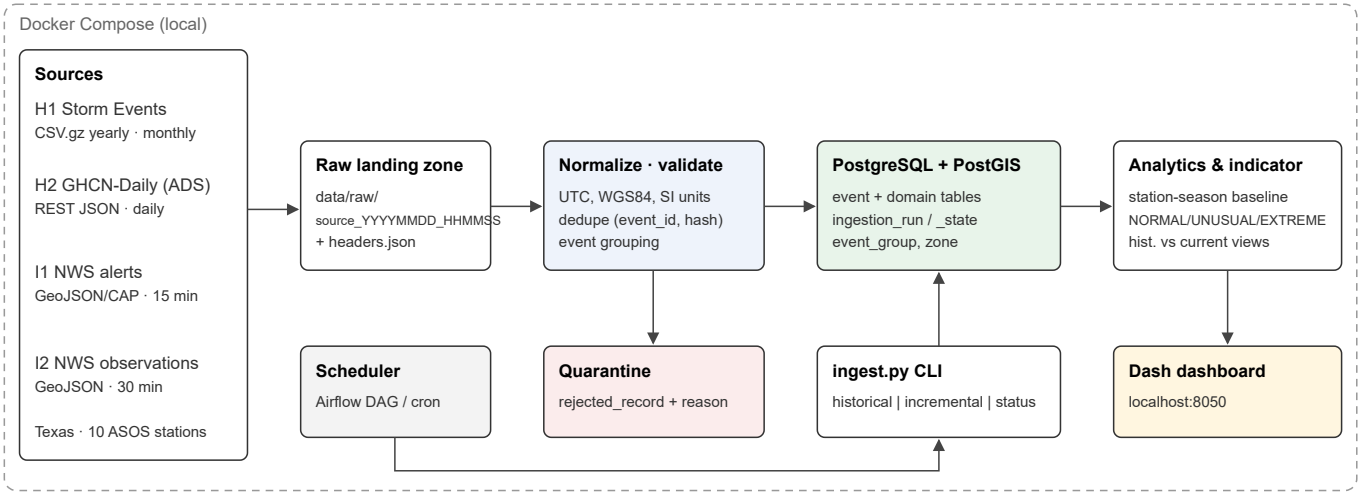


Figure 4. Reference architecture adapted to Team 5. The scheduler triggers `ingest.py --mode incremental`; the dashboard reads only from the database and its analytical views.

15. Repository

The repository follows the structure required by the workshop: `src/` (sources, normalization, storage), `ingest.py` (CLI stub with the final arguments), `config/sources.yaml` (one entry per source with URL, format, schedule and watermark field), `data/raw/` (git-ignored landing zone), `sql/` (draft schema), `dashboard/`, `tests/` and `tests/fixtures/` (recorded real responses, 17 passing tests), `notebooks/01_source_profiling.ipynb` (executed), `docs/` (this report, ingestion contract, source status log), `workshop_1/` (mapping.md, samples, scripts, data-quality table), `.env.example`, `.gitignore` and `README.md`. API keys are not needed by any of the four sources; the NWS User-Agent contact lives only in `.env`.

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[3] NOAA NCEI. Access Data Service API documentation. https://www.ncei.noaa.gov/support/access-data-service-api-user-documentation

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[5] OASIS. Common Alerting Protocol Version 1.2 (2010).

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